

CTR-GCN Person-Track Continuity

Formal Experimental Report

NTU RGB+D 120 · Interaction-26 · Cross-Subject
Global, Temporal, Localized and Sliding Localized Person Swap

11,660

Test samples

26

Two-person interaction classes

64 frames

Sampled model input

Principal findings

The trained CTR-GCN pipeline shows two distinct failure patterns: substantial Joint-stream sensitivity to arbitrary whole-clip person-slot order, and much stronger sensitivity to within-sequence person-track disruption. With duration and boundary count controlled, vulnerability rises toward the later sampled-input region and then falls at the final sampled window.

–8.76 pp

Joint Top-1 under Global Swap

–25.75 pp

Fusion Top-1 at W10 [40,56)

34.09%

Fusion prediction flips at W10

Evidence-led conclusions

- Bone is far more stable than Joint under Global Swap; fixed score addition partially mitigates the Joint failure.
- Localized Early/Middle/Late Fusion degradation is –9.53/–20.06/–25.75 pp, resolving the ambiguity of the suffix experiment.
- Joint reaches its largest accuracy drop earlier (center 40) than Bone and Fusion (center 48).
- The evidence motivates a minimal person-slot symmetry intervention before adding a larger interaction module.

Scientific scope. The experiments diagnose sensitivity to person-track continuity disruption. They do not demonstrate human identity recognition, and the temporal interventions can introduce artificial spatial, trajectory, velocity and motion discontinuities.

Research question and experimental scope

Two-person skeleton action recognition receives an explicit two-slot representation. The central question is whether predictions depend on arbitrary slot order or on maintaining a consistent assignment of each observed skeleton to a slot over time.

Preserved by the intervention

- The two skeletons present at every frame.
- Coordinates, frame order, labels and sample identity.
- The unordered per-frame pair of people.
- The trained checkpoint and test pipeline.

Changed by the intervention

- The assignment of skeletons to Person-0 / Person-1 slots.
- For partial swaps, continuity of each slot through time.
- Potentially, trajectory and velocity continuity at swap boundaries.

Why accuracy alone is insufficient

Aggregate accuracy can hide substantial output instability: correct→wrong and wrong→correct transitions can partially cancel. The study therefore reports Top-1, Top-5, their percentage-point changes, prediction flip rate, total variation (TV) between clean and perturbed probability distributions, and four correctness transitions.

Temporal coordinate. A window location refers to the sampled 64-frame model input after deterministic test preprocessing. It is not necessarily the same absolute time in every original NTU video.

CTR-GCN baseline

The diagnostic experiments use the completed two-stream baseline without retraining. Model input is **[B, 3, 64, 25, 2]**. Joint and Bone were trained independently; Fusion is a fixed score operation, not a learned third model.

Output	Selected epoch	Test Top-1	Test Top-5	Definition
Joint	60	88.49%	98.37%	Coordinates
Bone	55	88.33%	98.20%	Runtime bone vectors
Fusion	—	91.11%	98.82%	$S_{\text{fusion}} = S_{\text{joint}} + S_{\text{bone}}$ (fixed score addition)

11,764

Train samples

1,308

Validation samples

11,660

Test samples

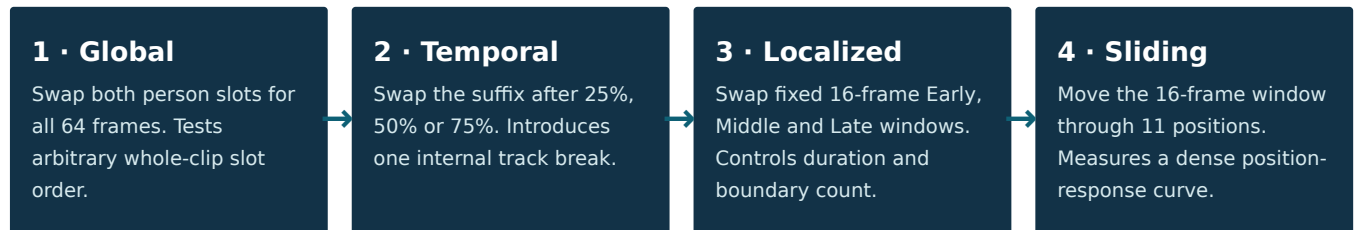
Test protocol

- Dataset: NTU RGB+D 120, Interaction-26 subset, Cross-Subject split.
- Deterministic test input length: 64 frames; two person slots.
- Checkpoint selection was completed before diagnostic test analysis.
- All four diagnostics reuse the same clean baseline scores and class labels.

Training observation. No person-slot swap augmentation was found in the saved training configuration. This observation is relevant context, not proof of the mechanism behind the measured response.

Four-stage experimental progression

The sequence was designed so that each stage addresses an ambiguity exposed by the preceding one.



Controlled semantics

- Global: all frames, no internal boundary.
- Temporal: variable swapped duration and one internal boundary.
- Localized/Sliding: 16 frames and exactly two internal boundaries.

- No checkpoint or fusion-weight search.
- No test-time randomness.
- No model retraining.
- Predictions, distributions and transitions independently validated from saved artifacts.

Interpretive logic

Global Swap separates whole-clip slot order from temporal track continuity. Temporal Swap establishes sensitivity to within-sequence disruption but confounds breakpoint and duration. Fixed Localized Swap removes that confound for three positions; Sliding Localized Swap tests whether the three-point pattern persists across the sampled timeline.

Global Person Swap

For every sample, the model-input person axis is exchanged for the entire clip: Person-0 ↔ Person-1 at every frame. Labels and the per-frame skeleton pair remain unchanged.

Stream	Condition	T1	Pert. T1	$\Delta T1$ pp	$\Delta T5$ pp	Flip	Mean TV
Joint	Global	88.49%	79.73%	-8.76	-2.04	17.40%	0.1980
Bone	Global	88.33%	87.68%	-0.64	-0.06	5.15%	0.0515
Fusion	Global	91.11%	89.16%	-1.96	-0.28	6.70%	0.0678

Observed response

CTR-GCN is not globally person-slot invariant. Joint loses 8.76 Top-1 points and flips 17.40% of predictions, whereas Bone loses only 0.64 points. Fusion reduces but does not remove the effect. The modality difference is an observed pattern; the experiment does not establish its cause.

Probability-scale note

Fusion TV in this table follows the formal J+B output. The Sliding plot later uses the separately validated normalized $0.5J+0.5B$ TV only to remove logit-scale ambiguity; its argmax is identical.

Fusion condition	C→C	C→W	W→C	W→W
Global	10,215	409	181	855

C→C: clean correct and perturbed correct; C→W: clean correct and perturbed wrong; W→C: clean wrong and perturbed correct; W→W: wrong in both conditions. Counts sum to 11,660.

Fixed-Breakpoint Temporal Swap

Frames before the breakpoint retain their slot assignment; all later frames exchange the two slots. The three conditions create one internal track discontinuity but differ in swapped duration.

Stream	Condition	T1	Pert. T1	$\Delta T1$ pp	$\Delta T5$ pp	Flip	Mean TV
Joint	25% / f16	88.49%	67.26%	-21.23	-8.34	31.05%	0.3522
Bone	25% / f16	88.33%	82.62%	-5.71	-1.78	13.38%	0.1386
Fusion	25% / f16	91.11%	83.10%	-8.01	-2.45	14.65%	0.1500
Joint	50% / f32	88.49%	51.11%	-37.38	-15.83	47.93%	0.4932
Bone	50% / f32	88.33%	76.74%	-11.59	-2.86	20.51%	0.2205
Fusion	50% / f32	91.11%	74.91%	-16.21	-4.13	23.77%	0.2451
Joint	75% / f48	88.49%	63.83%	-24.66	-8.94	35.11%	0.3708
Bone	75% / f48	88.33%	76.05%	-12.27	-3.22	21.25%	0.2221
Fusion	75% / f48	91.11%	78.70%	-12.42	-2.74	19.80%	0.2027

Interpretation. Fusion degradation is strongest at the 50% breakpoint (-16.21 pp), while Joint falls by 37.38 pp. This does not establish a middle-region peak: the 25%, 50% and 75% breakpoints swap 48, 32 and 16 suffix frames respectively.

All temporal conditions produce a larger Fusion response than Global Swap. This establishes sensitivity to within-sequence track disruption and motivates the fixed-duration Localized experiment.

Fixed 16-frame Localized Swap

Early [8,24), Middle [24,40) and Late [40,56) each exchange exactly 16 frames and introduce exactly two internal boundaries. Only temporal position changes.

Stream	Condition	T1	Pert. T1	$\Delta T1$ pp	$\Delta T5$ pp	Flip	Mean TV
Joint	Early [8,24)	88.49%	71.74%	-16.75	-6.92	25.86%	0.2952
Joint	Middle [24,40)	88.49%	54.61%	-33.88	-16.69	44.64%	0.4687
Joint	Late [40,56)	88.49%	51.79%	-36.70	-17.75	47.68%	0.4907
Bone	Early [8,24)	88.33%	77.71%	-10.62	-3.16	18.97%	0.2031
Bone	Middle [24,40)	88.33%	69.50%	-18.83	-6.05	28.32%	0.2971
Bone	Late [40,56)	88.33%	62.57%	-25.75	-10.71	35.91%	0.3620
Fusion	Early [8,24)	91.11%	81.59%	-9.53	-3.13	16.10%	0.1661
Fusion	Middle [24,40)	91.11%	71.05%	-20.06	-7.56	27.78%	0.2827
Fusion	Late [40,56)	91.11%	65.37%	-25.75	-10.62	34.09%	0.3446

-9.53 pp

Fusion Early $\Delta T_{\text{Top-1}}$

-20.06 pp

Fusion Middle $\Delta T_{\text{Top-1}}$

-25.75 pp

Fusion Late $\Delta T_{\text{Top-1}}$

Late > Middle > Early for Fusion degradation, flip rate and mean TV. Because duration and boundary count are fixed, the result supports a genuine temporal-position dependence within the sampled model input. It also resolves the ambiguity of the suffix experiment: its apparent peak cannot be read as an intrinsic middle-region maximum.

The intervention remains artificial. Equal duration does not make the swap a pure identity manipulation, and it does not imply that window length is irrelevant in other settings.

Sliding Localized Swap: 11-point response

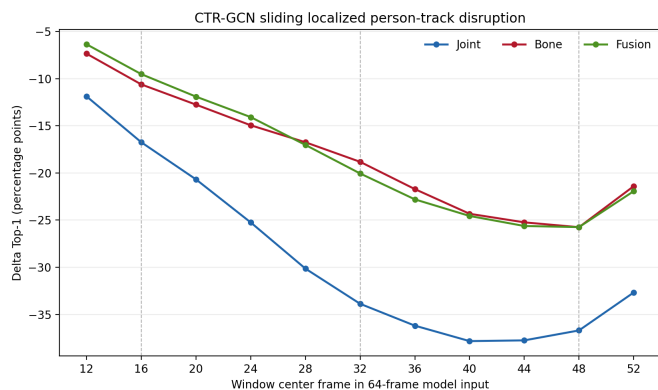


Figure 1. Top-1 degradation across window centers.

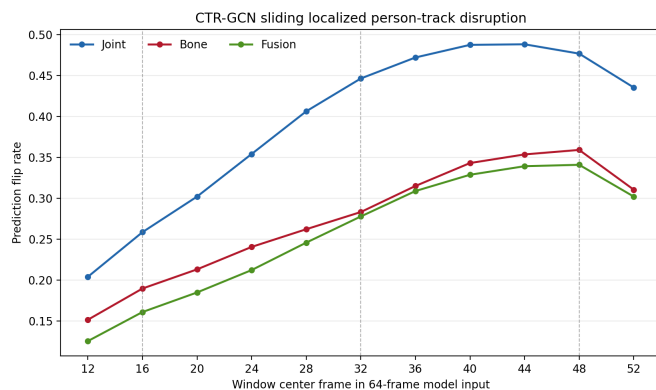
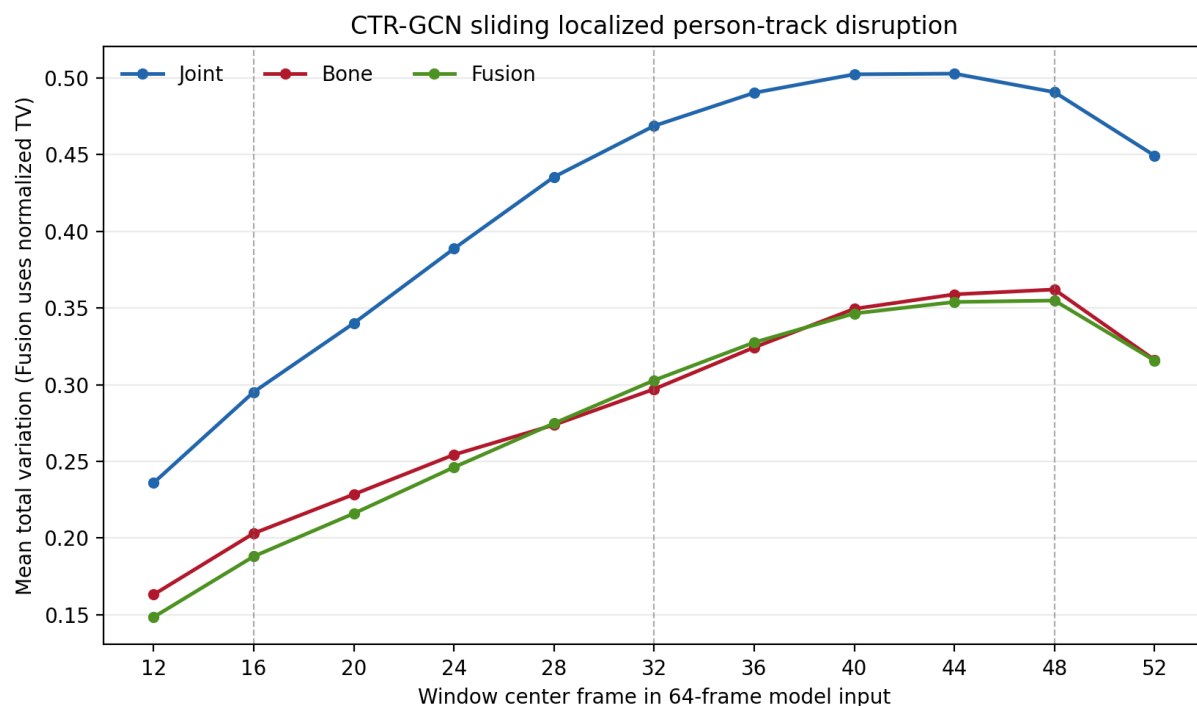


Figure 2. Prediction flip rate.

Figure 3. Mean total variation. For CTR-GCN, Fusion uses normalized $0.5J+0.5B$ probabilities; Joint and Bone use their native logits.

The curves rise broadly toward the mid-late and late sampled-input regions, but are not strictly monotonic. W02, W06 and W10 reproduce the earlier Early/Middle/Late checkpoints; all three regression checks passed.

Joint, Bone and Fusion response

Stream	Early avg $ \Delta T1 $	Midpoint	Late avg $ \Delta T1 $	Max Δ window	Max flip	Max TV
Joint	20.94	33.88	36.23	W08 / c40	48.82%	0.5028
Bone	12.49	18.83	23.69	W10 / c48	35.91%	0.3620
Fusion	11.78	20.06	24.14	W10 / c48	34.09%	0.3548

Joint is the most sensitive stream across the curve and has an earlier/broader peak: its maximum accuracy drop occurs at center 40, while its flip and TV maxima occur at center 44. Bone and Fusion peak at center 48. Fusion substantially improves clean accuracy and is more robust than Joint, but does not eliminate the failure.

Fusion 11-point values

ID	Window	Center	$\Delta T1$ pp	Flip	Mean TV
W01	[4,20)	12	-6.36	12.55%	0.1485
W02	[8,24)	16	-9.53	16.10%	0.1881
W03	[12,28)	20	-11.91	18.49%	0.2160
W04	[16,32)	24	-14.08	21.23%	0.2461
W05	[20,36)	28	-17.03	24.58%	0.2748
W06	[24,40)	32	-20.06	27.78%	0.3028
W07	[28,44)	36	-22.80	30.88%	0.3275
W08	[32,48)	40	-24.56	32.87%	0.3464
W09	[36,52)	44	-25.63	33.92%	0.3539
W10	[40,56)	48	-25.75	34.09%	0.3548
W11	[44,60)	52	-21.94	30.21%	0.3157

Early average uses centers <32 ; midpoint is center 32; late average uses centers >32 . Spearman position trends are exploratory summaries over only 11 measured positions, not causal tests.

Where sensitivity concentrates

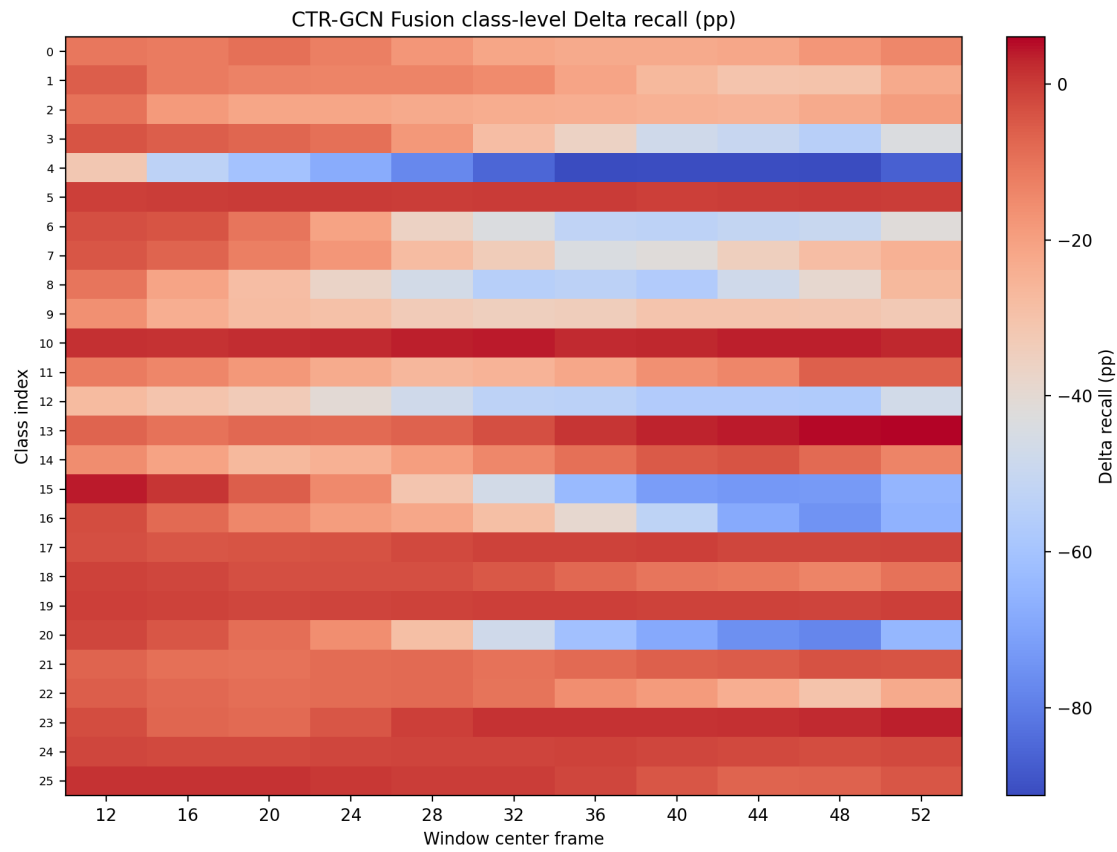


Figure 4. Fusion recall change by class and sliding window. The complete 26×11 values remain in the experiment artifacts; the report presents only selected summaries.

Largest late-minus-early concentration

Class	Late-early drop	W10 drop	W10 flip
Class 15	+57.95 pp	73.04 pp	95.48%
Class 20	+57.57 pp	77.95 pp	81.08%
Class 16	+46.82 pp	74.61 pp	78.96%
Class 3	+37.46 pp	54.71 pp	59.78%
Class 6	+34.57 pp	49.64 pp	56.16%

Observed failure patterns

- Whole-clip slot-order sensitivity: major in Joint.
- Strong within-sequence track-disruption sensitivity.
- Temporal-position-dependent vulnerability, concentrated later.
- Different Joint and Bone response shapes.
- Substantial class-dependent variation.

Relatively stable on mean recall drop: Class 5, Class 19, Class 24, Class 17.

CTR-GCN class indices are retained because the exact index-to-action-name mapping was not independently verified in its artifacts. Class rankings describe concentration of measured response; they do not identify a semantic cause.

Evidence-to-design reasoning

Experimental evidence	Observed failure pattern	Design requirement	Candidate modification	Support / complexity / risk
Joint: -8.76 pp under Global Swap	Arbitrary person-slot dependence	Permutation-compatible pair handling	Swap-consistency loss or symmetric pair aggregation	High / low / low
Large localized drops in both streams	Track disruption changes recognition	Preserve person-specific temporal features	Delay person pooling; retain $h_A(t)$, $h_B(t)$	High / medium / medium
Late-region peak; non-monotonic curve	Narrow temporal reliance	Multi-scale interaction aggregation	Lightweight temporal pyramid or phase-aware pooling	Medium / medium / medium
Joint/Bone robustness differs	Fixed score sum leaves complementarity unused	Structured but restrained fusion	Confidence-gated or validation-fixed learned fusion	Medium / medium / medium

Observation

Measured global, temporal and class-dependent response

Design implication

Preserve symmetry or temporal pairing where the evidence demands it

Minimal modification

Change one representation or loss component and ablate it independently

Priority. The first intervention should target person-slot symmetry because the Global Joint failure is large, specific, and addressable with an almost parameter-free ablation. Larger temporal-relation modules should follow only if this does not damage clean recognition.

Explicit cross-person relations—relative joint displacement, distances, relative velocity, relative bone geometry or learned pair tokens—are plausible later directions. They remain hypotheses until clean and diagnostic ablations show a benefit.

Swap-consistency regularisation for the Joint stream

During training, evaluate each Joint sample in original and whole-clip swapped order and add a small symmetric consistency term between their logits or probability distributions, while retaining the ordinary classification loss. This introduces no inference-time parameters and directly targets the measured Global failure.

Required ablation

Baseline; swap augmentation only; consistency only; augmentation + consistency. Keep Bone and fixed J+B fusion unchanged so the effect is attributable.

Baseline comparison

Use the same data split, checkpoint-selection procedure and clean evaluation. Do not choose the modification from Test perturbation results.

Success criteria

- Maintain or improve clean Test Top-1; report Macro-F1 when implemented.
- Report per-class recall, parameter count and computational overhead.
- Reduce the model-relevant Global and selected Localized/Sliding sensitivity.
- Verify that robustness is not obtained by suppressing useful motion evidence.

Limitations

The study uses one trained baseline line and artificial slot interventions. It cannot separate identity, spatial jumps, trajectory changes, velocity discontinuities and other motion inconsistencies. Temporal positions are defined only in the sampled 64-frame input. Exploratory position trends and class rankings require confirmation on additional seeds and, ideally, other protocols.

Conclusion

CTR-GCN exhibits both arbitrary slot-order sensitivity—concentrated in Joint—and strong temporal continuity dependence across both streams. The controlled Localized and Sliding experiments identify a later-region vulnerability without establishing a causal mechanism. The recommended next step is deliberately minimal, testable and reversible.